

⚡ SCIENCE FAIR 2026 • AI INNOVATION

Can AI Predict **Student Failure...** **Before Exams Even Start?**

Traditional report cards tell parents what went wrong after final marks are locked. We engineered an intelligent, early-warning ML mentorship platform to intervene proactively.

81.4%**Prediction Accuracy**

Gradient Boosting Engine

12 Core**Student Indicators**

Attendance, Habits & Study

1-Page**Instant Report Card**

Official School Seal

**Live Cloud Application Deployed on Vercel**<https://eduvision-ai-kappa.vercel.app>**TESTED & LIVE →**

⚠ THE SYSTEMIC FLAW

Why Traditional School Report Cards Are **Fatally Broken**:

Teachers manage 40+ students per classroom. Spotting silent behavioral dropouts before final exams is nearly impossible without intelligent data.

✗ 01. Post-Mortem Analysis

Interventions Happen Too Late

Corrective tutoring only begins after a student fails final exams, leaving zero recovery time.

✗ 02. Invisible Behavioral Clues

Silent Warning Signs Ignored

Micro-absences and subtle study drops silently degrade comprehension weeks before any test paper is written.

✗ 03. Vague & Generic Feedback

"Study Harder" Is Not A Strategy

Students get no concrete targets. They need exact study hour benchmarks and focused intervention roadmaps.

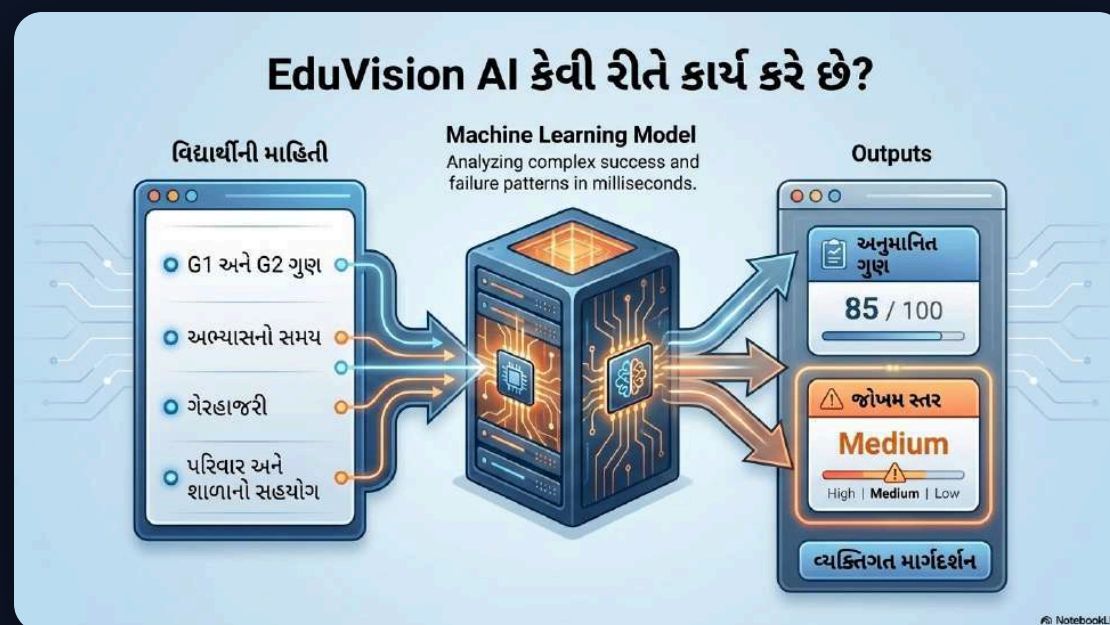
💡 04. The AI Solution

Proactive Early Intervention

Machine learning flags at-risk students months before exams with tailored step-by-step recovery plans.

Meet **EduVision AI**: Proactive Student Mentorship

Trained on 395 real-world student longitudinal profiles (UCI Benchmark), predicting final grades (G3) with over 81% precision.



12 Core Inputs

Captures study hours, absences, family support, internet access, and past marks.

ML Pipeline

Scikit-learn ColumnTransformer with robust feature scaling and encoding.

Actionable Output

Instant predicted score out of 20, HUD risk gauge, and customized guidance.

 ALGORITHMIC SUPERIORITY

Why **Gradient Boosting** Won The Tournament:

We trained, tuned, and evaluated three distinct regression architectures. Gradient Boosting systematically outperformed all others:

Linear Regression (Baseline)

MAE: 1.4429 R² Score: 0.7690

Random Forest (Ensemble)

MAE: 1.2061 R² Score: 0.8132

 **Gradient Boosting** **CHAMPION**

MAE: 1.1804 (Lowest) R²: 0.8138 (81.4%)

Key Technical Takeaway:

Gradient Boosting iteratively minimizes residuals from previous decision trees. It handled non-linear behavioral traits like absenteeism and past failures far better than conventional linear or bagging models.

The Zero-Collision HUD Gauge Meter

Engineered for high-visibility kiosk projection at the Science Fair, translating raw mathematical regressions into an intuitive cybernetic visual meter.

🎯 01. Dynamic SVG Arc

Zero Overlapping Needle Glitches

Active neon arc illuminates smoothly to the student's exact predicted percentage with crisp visual feedback.

🌐 02. Gujarati Localization

Accessible to Rural Students

Instant toggle between English & Gujarati using modern regional typography (Anek Gujarati & Noto Sans).

🚦 03. 3-Tier Risk Zones

Immediate Triage Classification

🔴 High (<50%) • 🟡 Mid (50-70%) • 🟢 Safe (>70%)

⚡ 04. Sub-50ms Response

Real-Time Fluid Interactivity

Zero lag during student interactions, providing immediate cause-and-effect visualization.



The "What-If" Academic Simulator

A score is not a final verdict—it is a roadmap. Students move interactive sliders to discover exactly what behavioral changes alter their future.



"What if I study 2 more hours?"

Instantly recalibrates:
+1.8 Marks Gain. Proves the tangible return on daily study routines.



"What if attendance hits 95%?"

Reduces failure probability: **Risk Drops**
● → ●. Highlights the power of classroom presence.



"What if parental support is High?"

Builds academic resilience: **Stabilizes Variance** and cushions test anxiety during exams.



 SCIENCE FAIR TAKE-HOME

Instant 1-Page Official Report Card

At our Science Fair kiosk, students and parents hit "Print Report" to walk away with a verified, single-page A4 physical certificate.

01. Zero-UI Print Portal

Pristine A4 Formatting

Isolated React Portal & strict CSS @media print rules strip out navbars, buttons, and background glows, ensuring a crisp document.

02. Verified School Endorsement

Official School Seal & Signatures

Includes M. M. Karodiya Primary School crest, official mentor signature block (Shri Manojbhai Parmar), and parent endorsement.

Comprehensive 12-Indicator Diagnostic

The report doesn't just show a number—it itemizes specific strengths, areas for growth, and personalized weekly study targets for parents.

 TRY IT LIVE TODAY

Experience **EduVision AI 2.0**

Fully deployed on Vercel with zero latency, complete Gujarati localization, and 100% open-source code on GitHub.

 LIVE INTERACTIVE WEB APP

<https://eduvision-ai-kappa.vercel.app>

TRY DEMO →

 FULL OPEN SOURCE REPOSITORY

<https://github.com/yashintelligence/eduvision-ai>

STAR REPO ★

Special thanks to **M. M. Karodiya Primary School** and project mentor **Shri Manojbhai Parmar**.